

Smart Irrigation

by Tom Ash

Water is pumped for miles, sometimes hundreds of miles, from its source to 90% of U.S. homes. This takes an enormous amount of energy; in California, it represents the single largest use of energy statewide. And water is heavy: 1 gallon weighs 8.3 lb. A typical U.S. household uses an average of 87,000 gallons, or 722,100 lb of water per year on landscaping alone. Using water efficiently, then, reduces the amount of energy needed to move water.

Fifty-eight percent of the water a typical home uses goes to landscaping. This keeps the garden green and healthy in some cases, but in others, much of that water is wasted. Public agency studies from Florida to Oregon show that installing a certified “smart” irrigation controller is the most direct way to avoid overwatering and save water outdoors.

What’s so smart about a smart controller? A smart controller hooks up to irrigation valves with the same valve wires as a conventional controller. It irrigates at whatever time of day or night you choose. You can turn it on manually to check the irrigation system, and you can set any station on a regular watering cycle to establish plants. But there the similarity between conventional and smart controllers ends—and a new age of water efficiency begins.

Let’s look at what Linda Leisner of Irvine, California, has to say. “My water use has come down 50% since I installed a smart controller,” says Linda. And she adds, “I think everyone should do this.”

What did Linda do? She installed WeatherTRAK, the most thoroughly tested smart controller on the market. WeatherTRAK calculates an efficient watering schedule for each station or zone in a garden. How this smart controller works is it



MICHAEL HARRISON

Landscaper Roger Waters explains the water-saving features of the WeatherTRAK system to homeowner Anne Digges.

receives daily local weather updates that it uses to determine each zone’s evapotranspiration (ET) rate—a combination of the soil’s and the plant’s evaporation rates in that weather. It uses this information to determine how much water the plants in that zone will need. In cool weather the smart controller might not water the plants for several days. In hot weather, the controller will water as long and as often as the plants need. It does all this automatically, so you don’t have to be a meteorologist or a horticultural expert to make it work.

WeatherTRAK was the first product to receive a perfect score on the Irrigation Association’s Smart Water Application Technology, or SWAT, test. In every public agency study, WeatherTRAK technology saved water and improved the health and appearance of plants. But what customers say they like most about smart controllers is that they’re so convenient. With a smart controller, you don’t need to set anything, change anything, or turn anything on or off as the weather changes. Go on vacation,

and your smart controller will not overwater or underwater your garden.

In Linda’s yard there is a wide range of plants—from turf to shade-loving plants to plants that require almost no water. There are zones of deep shade and zones of all-day full sun. For every station or zone, a series of simple questions appears on the smart controller’s monitor, questions about type of soil, type of plant, sun or shade, slope, and type of sprinkler. When Linda sets up the controller, the monitor might display Plant Type, for example. Linda turns the knob to select Turf, Shrubs, Flowers, or Native Plants, as the case may be, and the controller calculates how many minutes and cycles to water to keep the plants healthy and to maintain maximum water efficiency. Daily weather updates enable the controller to adjust the watering schedule as needed.

With a 50% reduction in her landscape water use, Linda Leisner is saving approximately 8 million kilowatts per year of the energy used to pump the water to her home from the water agency.



HYDROPOINT

The WeatherTRAK ET plus controller automatically adjusts watering based on daily, local weather conditions.

Today, builders throughout the West are specifying WeatherTRAK smart controllers to help save water, reduce urban runoff, and eliminate water damage to homes caused by overwatered landscapes. In some communities where water runoff is polluting beaches, bays, and waterways, cities are providing rebates and free

smart controllers to residents. Newport Beach, California, implemented such a program recently, and residents lined up to have the WeatherTRAK installed, eager to eliminate polluted runoff from landscapes and help the local environment. Ron Vanderhoff, nursery manager of Rogers Gardens, the largest retail nursery in the United States, says,

“Our customers want a great garden, but they also want to do the right thing for the environment. We suggest starting with a really smart controller. We have installed WeatherTRAK in our demonstration garden and in our maintenance accounts. It works.”

With summer coming, water use—and energy use—tend to climb. But with a smart controller, they don’t have to climb too high.



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For more information:

Find out more about smart controllers at www.weathertrak.com. At this site you can view public agency studies, news reports, and a film clip featuring a smart controller from ABC TV’s Good Morning America.

To find out more about the Irrigation Association’s SWAT test, visit www.irrigation.org.